

2023 Data Mining

HW2

Task introduction

- Income prediction
 - Implement logistic regression **using only numpy** to predict the income is over than 50K or not.
 - pandas, csv and matplotlib are available for data analysis and pre-processing



- Requirement
 - Upload your submission to Kaggle
 - Submit a report and your source code to E3

- Deadline is 5/9 23:59, no late submission



**Adult UCI
Census Income
Project**

Machine Learning

Dataset

UCI Adult Income Data Set.

- train.csv
 - Attributes of Adults for training. Randomly selected 80% from whole dataset.
 - [link](#)
- test_X.csv
 - Attributes of Adults for testing, which is remaining 20% of dataset.
 - Please use the features to predict income is over 50K or not.
 - [link](#)

Training Data

age	workclass	fnlwgt	education	educational-num	marital-status	occupation	relationship	race	gender	capital-gain	capital-loss	hours-per-week	native-country	income
31	Private	137991	Some-college	10	Married-AF-spouse	Sales	Wife	White	Female	0	0	20	United-States	<=50K
58	Private	51662	10th	6	Married-civ-spouse	Other-service	Wife	White	Female	0	0	8	United-States	<=50K
39	Private	56962	Some-college	10	Married-civ-spouse	Exec-managerial	Wife	White	Female	0	0	43	United-States	>50K
33	Private	169886	12th	8	Separated	Other-service	Unmarried	White	Female	0	0	50	Dominican-Republic	<=50K
68	Private	168794	Preschool	1	Never-married	Machine-op-inspct	Not-in-family	White	Male	0	0	10	United-States	<=50K
30	Private	243841	HS-grad	9	Divorced	Other-service	Other-relative	Asian-Pac-Islander	Female	0	0	40	South	<=50K
34	Local-gov	211239	Bachelors	13	Divorced	Prof-specialty	Unmarried	White	Female	6497	0	40	United-States	<=50K
27	Private	169815	Bachelors	13	Never-married	Sales	Own-child	White	Male	0	0	40	United-States	<=50K
53	Private	95519	HS-grad	9	Married-civ-spouse	Prof-specialty	Husband	White	Male	0	0	42	United-States	>50K
18	Private	194059	12th	8	Never-married	Adm-clerical	Own-child	White	Male	0	0	40	United-States	<=50K
72	Private	74141	9th	5	Married-civ-spouse	Exec-managerial	Wife	Asian-Pac-Islander	Female	0	0	48	United-States	>50K
59	Self-emp-inc	171355	Masters	14	Divorced	Prof-specialty	Unmarried	White	Male	0	0	55	United-States	<=50K
32	Private	38797	Assoc-voc	11	Never-married	Craft-repair	Not-in-family	White	Male	0	0	40	United-States	<=50K
60	Private	80574	HS-grad	9	Divorced	Other-service	Unmarried	White	Female	0	0	38	United-States	<=50K
23	Private	202084	HS-grad	9	Never-married	Other-service	Unmarried	Black	Female	0	0	40	United-States	<=50K
51	Private	186303	Bachelors	13	Married-civ-spouse	Sales	Husband	White	Male	0	0	40	United-States	>50K
47	Self-emp-not-inc	265097	HS-grad	9	Married-civ-spouse	Transport-moving	Husband	White	Male	0	0	45	United-States	>50K
27	State-gov	261278	Prof-school	15	Never-married	Prof-specialty	Not-in-family	Black	Female	0	0	80	?	<=50K
20	Private	82777	HS-grad	9	Married-civ-spouse	Sales	Own-child	White	Male	0	0	40	United-States	<=50K
59	Self-emp-not-inc	32537	HS-grad	9	Married-civ-spouse	Farming-fishing	Husband	White	Male	0	0	40	United-States	>50K
23	Private	117789	Bachelors	13	Never-married	Prof-specialty	Not-in-family	White	Female	0	0	60	United-States	<=50K
25	Private	372728	Bachelors	13	Never-married	Other-service	Not-in-family	Black	Female	0	0	24	Jamaica	<=50K
62	Private	49424	Bachelors	13	Married-civ-spouse	Prof-specialty	Husband	White	Male	5178	0	40	United-States	>50K
62	Private	138621	Assoc-voc	11	Separated	Priv-house-serv	Not-in-family	Black	Female	0	0	20	United-States	<=50K
22	Private	203182	Some-college	10	Separated	Sales	Unmarried	White	Female	0	0	43	United-States	<=50K

? represented as an unsure value

Testing Data

age	workclass	fnlwgt	education	educational-num	marital-status	occupation	relationship	race	gender	capital-gain	capital-loss	hours-per-week	native-country
42	State-gov	172307	Some-college	10	Married-civ-spouse	Exec-managerial	Husband	White	Male	0	0	45	United-States
39	Private	240468	HS-grad	9	Never-married	Exec-managerial	Not-in-family	White	Male	0	0	42	United-States
44	Private	151089	Masters	14	Married-civ-spouse	Prof-specialty	Husband	White	Male	0	1887	60	United-States
38	Private	269318	Bachelors	13	Married-civ-spouse	Adm-clerical	Husband	White	Male	5178	0	50	United-States
31	Private	161153	HS-grad	9	Married-civ-spouse	Machine-op-inspct	Husband	White	Male	0	0	45	United-States
19	Private	253529	12th	8	Never-married	Adm-clerical	Own-child	White	Male	0	0	9	United-States
34	Self-emp-not-inc	156532	7th-8th	4	Divorced	Farming-fishing	Not-in-family	White	Male	0	0	50	United-States
23	Private	209955	Some-college	10	Never-married	Sales	Own-child	White	Male	0	0	40	United-States
46	Private	235334	Masters	14	Married-civ-spouse	Exec-managerial	Husband	Black	Male	99999	0	60	United-States
41	Private	279914	Some-college	10	Married-civ-spouse	Adm-clerical	Husband	White	Male	0	0	45	United-States
54	Private	249322	Bachelors	13	Married-civ-spouse	Sales	Husband	White	Male	7688	0	50	United-States
37	Private	321943	Prof-school	15	Married-civ-spouse	Prof-specialty	Husband	White	Male	0	0	40	United-States
51	Private	101722	7th-8th	4	Married-civ-spouse	Exec-managerial	Husband	Amer-Indian-Eskimo	Male	3908	0	47	United-States
30	Private	140612	HS-grad	9	Married-civ-spouse	Craft-repair	Husband	White	Male	0	2051	40	United-States
62	Self-emp-not-inc	320376	HS-grad	9	Married-civ-spouse	Sales	Wife	White	Female	0	0	48	United-States
26	Private	58426	HS-grad	9	Married-civ-spouse	Prof-specialty	Husband	White	Male	0	0	50	United-States
34	Local-gov	206707	Some-college	10	Married-civ-spouse	Protective-serv	Husband	White	Male	0	0	60	United-States

Same attributes but no income value, you need to predict them.

Attributes Description

- age: continuous.
- workclass: Private, Self-emp-not-inc, Self-emp-inc, Federal-gov, Local-gov, State-gov, Without-pay, Never-worked.
- fnlwgt: continuous.
 - The number of people the census takers believe that observation represents.
 - Don't ask TA because they don't know it, either.
- education: Bachelors, Some-college, 11th, HS-grad, Prof-school, Assoc-acdm, Assoc-voc, 9th, 7th-8th, 12th, Masters, 1st-4th, 10th, Doctorate, 5th-6th, Preschool.
- education-num: continuous.

Attributes Description

- marital-status: Married-civ-spouse, Divorced, Never-married, Separated, Widowed, Married-spouse-absent, Married-AF-spouse.
- occupation: Tech-support, Craft-repair, Other-service, Sales, Exec-managerial, Prof-specialty, Handlers-cleaners, Machine-op-inspct, Adm-clerical, Farming-fishing, Transport-moving, Priv-house-serv, Protective-serv, Armed-Forces.
- relationship: Wife, Own-child, Husband, Not-in-family, Other-relative, Unmarried.
- race: White, Asian-Pac-Islander, Amer-Indian-Eskimo, Other, Black.
- gender: Female, Male.

Attributes Description

- capital-gain: continuous.
- capital-loss: continuous.
- hours-per-week: continuous.
- native-country: United-States, Cambodia, England, Puerto-Rico, Canada, Germany, Outlying-US(Guam-USVI-etc), India, Japan, Greece, South, China, Cuba, Iran, Honduras, Philippines, Italy, Poland, Jamaica, Vietnam, Mexico, Portugal, Ireland, France, Dominican-Republic, Laos, Ecuador, Taiwan, Haiti, Columbia, Hungary, Guatemala, Nicaragua, Scotland, Thailand, Yugoslavia, El-Salvador, Trinidad&Tobago, Peru, Hong, Holand-Netherlands.

Kaggle Submission

- [Kaggle link](#)
- Display team name : <student ID>
- Submission format
 - A 9970*2 .csv file, index start from 0. 1 represent income > 50K, and 0 is <= 50K.
 - Column name must be **id** and **income**.
 - [sample submission](#)
- There are one simple bassline and one strong bassline, beat them to get the higher score.

id	Income
0	0
1	0
2	0
3	1
4	0
5	0
6	0
7	0
8	1
9	0

Kaggle Submission

- You can submit at most 5 times each day.
- The scoring metric is **Acc**.
- You can choose 2 of the submissions to be considered for the private leaderboard, or will otherwise default to the best public scoring submissions.

Report Submission

Answer the following 3 questions:

1. How do you select features for your model input, what preprocessing did you do ?
2. Discuss which attribute has the greatest impact on income prediction accuracy, visualize the result and explain it.
3. Discuss the impact of regularization on income prediction accuracy.

Please answer the question in detail !

Grading policy

- Kaggle (70%)
 - Basic score:
 - Over strong baseline : 55
 - Over simple bassline : 40
 - Under simple baseline : 25
 - Ranking score:
 - $15 - (15/N) * (\text{ranking} - 1)$, N =numbers of people
 - 30% public leaderboard 70% private leaderboard
- Source code and report (30%)
 - 10% for each quesiton

E3 Submission

Submit your source code and report to E3 before 5/9 23:59.

No late submission !

- Format

- source code : HW2_<student ID>.py or HW2_<student ID>.ipynb
- report : HW2_<student ID>.pdf

If you have any question about HW2, please feel free to contact with TA: GUAN-YI HO
through email joy861106.cs11@nycu.edu.tw

Have Fun

